25901274 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEFTQ200 - FOOD PACKAGING TECHNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer type

Answer any **10** questions

Each question carries **2** marks

1. State any two advantages of glass containers for packaging food products. [U] / [CO1]
2. How does corrosion affect metal cans used for food storage? [A] / [CO1]
3. State two uses of paper in food packaging. [U] / [CO1]
4. Identify one packaging material with high tear strength. [K] / [CO2]
5. State one difference between MAP and CAP. [K] / [CO3]
6. Explain how biodegradable packaging differs from conventional plastic packaging. [U] / [CO3]
7. What is meant by plastic migration in food packaging? [U] / [CO1]
8. List one application of stretch wrapping. [K] / [CO3]
9. Define food packaging. [K] / [CO2, CO5]
10. Define intelligent packaging. [K] / [CO3]

11. Name any two types of colored glass containers used in food packaging. [A] / [CO1]

12. Name one method used to evaluate grease resistance of packaging materials. [K] / [CO5]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **6** questions

Each question carries **5** marks

13. Explain how retort pouches compare with glass jars in terms of safety and durability. [U] / [CO1]

14. Explain the need for food packaging. [E] / [CO2]

15. Explain the importance of hermetic sealing in canned food products. [U] / [CO5]

16. Explain the shelf-life attributes of foods packaged in glass containers. [K] / [CO1]

17. Describe the role of retort pouches in extending the shelf life of food products. [U] / [CO5]

18. Write a note on thickness testing of metal sheets used in cans. [U] / [CO2]

19. Define edible films and explain their importance in food packaging. [U] / [CO1]

20. Discuss the solubility tests used for identifying common plastics. [U] / [CO5]

[5x6 = 30]

Part C

Long Essay Questions

Answer any **2** questions

Each question carries **10** marks

21. Explain the working principle of aseptic packaging with neat diagrams. [U] / [CO3]

22. Explain the demerits of plastics in food packaging. Discuss environmental concerns, food safety issues, and limitations in certain applications. [U] / [CO1]

23. Define primary, secondary and tertiary packaging and explain with examples. [K] / [CO1]

24. Discuss the FSSAI regulations related to food packaging and labelling. [K] / [CO2]

[10x2 = 20]